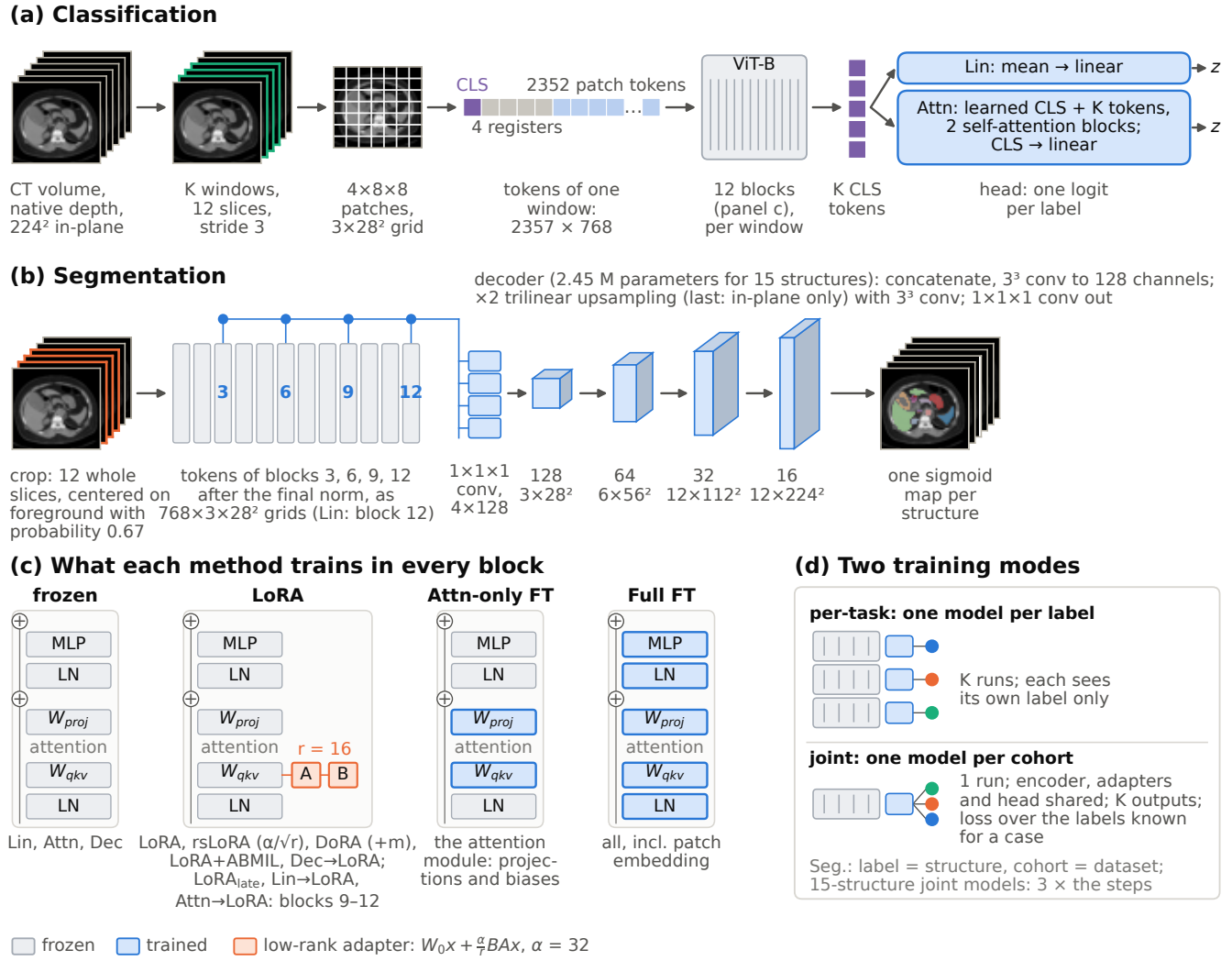

(a) Classification
CLS
2352 patch tokens
4 registers
ViT-B
Lin: mean → linear
Attn: learned CLS + K tokens, 2 self-attention blocks; CLS → linear
z
z
CT volume, native depth, 224² in-plane
K windows, 12 slices, stride 3
4×8×8 patches, 3×28² grid
tokens of one window: 2357 × 768
12 blocks (panel c), per window
K CLS tokens
head: one logit per label
(b) Segmentation
decoder (2.45 M parameters for 15 structures): concatenate, 3³ conv to 128 channels; ×2 trilinear upsampling (last: in-plane only) with 3³ conv; 1×1×1 conv out
3
6
9
12
crop: 12 whole slices, centered on foreground with probability 0.67
tokens of blocks 3, 6, 9, 12 after the final norm, as 768×3×28² grids (Lin: block 12)
1×1×1 conv, 4×128
128 3×28²
64 6×56²
32 12×112²
16 12×224²
one sigmoid map per structure
(c) What each method trains in every block
frozen
LoRA
Attn-only FT
Full FT
MLP
LN
W_{proj}
attention
W_{qkv}
LN
r = 16
A
B
Lin, Attn, Dec
LoRA, rsLoRA (α/√r), DoRA (+m), LoRA+ABMIL, Dec→LoRA; $LoRA_{late}$, Lin→LoRA, Attn→LoRA: blocks 9–12
the attention module: projections and biases
all, incl. patch embedding
(d) Two training modes
per-task: one model per label
K runs; each sees its own label only
joint: one model per cohort
1 run; encoder, adapters and head shared; K outputs; loss over the labels known for a case
Seg.: label = structure, cohort = dataset; 15-structure joint models: 3 × the steps
frozen
trained
low-rank adapter: $W_0x + \frac{\alpha}{r}BAx$, $\alpha = 32$